

TITLE

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1 Introduction

1.1 Collostructional analysis and learner corpora

1.2 Formulaicity in learner language

- RQ1. How are type II conditionals dispersed across learner writing tasks present in our analyzed data?
- RQ2. Do learners at higher proficiency levels favor different verbs in the protasis of type II conditionals than beginner learners?
- RQ3. Does the verbal slot in the protasis of type II conditionals provide evidence for a transition from more formulaic to more productive use of the construction?

2 Methods

2.1 Data: The EFCAMDAT

The EF-Cambridge Open Language Database (EFCAMDAT), first presented by Geertzen et al. (2013), is a large-scale learner corpus containing writings from the online English school *Englishtown*. In their courses, each level corresponds to eight pedagogical units, after each of which students receive a prompt for a writing task that they must complete in order to advance. The corpus comprises responses to these tasks in their original and corrected forms, along with metainformation about the text and the author (learner ID, country of origin, grade given by a teacher, etc.). An advantage of the EFCAMDAT, beyond being generally available for researchers at all stages, is the fact that *Englishtown* levels are aligned with

the CEFR, making it possible to do a broad range of descriptive analyses of L2 English at globally understood proficiency levels.¹

The version of EFCAMDAT used for this study contained 620,206 learner writings in total. However, it was unlikely for second conditionals to be elicited by most prompts; therefore, after qualitatively inspecting the task prompts, I decided on a subset of 20 tasks whose responses were most likely to contain the construction. The selected units and corresponding writing topics are outlined in Table 1, totaling 33,496 student writings and 4,402,652 words.

CEFR level	EF level:Unit	Topic ID	Words	Topic
A2	6:6	46	290,594	Writing an email of advice
B1	8:3	59	474,441	Making a ‘to do’ list of your dreams
B1	9:3	67	354,123	Making a business proposal
B1	9:4	68	284,752	Signing a waiver to go skydiving
B2	10:1	73	797,218	Helping a friend find a job
B2	10:2	74	492,941	Doing a survey about discrimination
B2	10:3	75	438,282	Requesting a bank loan
B2	10:5	77	313,135	Finding a home for a wealthy client
B2	11:2	82	291,016	Helping a coworker deal with a phobia
B2	11:8	88	24,119	Improving your study skills
C1	13:1	97	244,337	Writing a campaign speech
C1	13:4	100	115,523	Giving advice about budgeting
C1	13:8	104	21,223	Reaching your potential
C1	14:2	106	41,642	Choosing a renewable energy source
C1	14:3	107	61,155	Writing a rejection letter
C1	14:4	108	55,637	Attending a seminar on stress reduction
C1	14:5	109	52,968	Talking a friend out of a risky action
C1	14:6	110	39,465	Applying for sponsorship
C1	15:7	119	5,182	Writing about future lifestyles
C1	15:8	120	4,899	Interpreting a prophecy

Table 1: Topics by Level and CEFR

2.2 Extracting conditionals

All corrected learner writings in the subcorpus were automatically parsed using Stanza (Qi et al., 2020). Using the corrected texts instead of the original ones is justified by the fact that

¹The availability of large-scale, CEFR-aligned corpora is more limited than one might expect. The most well-known English learner corpus is the Cambridge Learner Corpus (CLC), whose access is highly restricted.

accuracy does not factor into this study: the goal is not to assess the grammatical proficiency of students, but rather to explore their lexical preferences in a specific slot in different levels. This gives us the ability to use the corrected texts that are present in the corpus, ensuring more robust results of automatic Natural Language Processing (NLP) tools.²

The parsed documents were scanned for type II conditionals using both Universal Dependency (UD) relations and part-of-speech (POS) information. Only sentences containing the word "if" as a conditional marker (headed by an *advcl* UD tag) and not as an interrogative marker headed by a *ccomp* UD tag, as in Figure 1. Then, the script ensures that the verb in the protasis is in the past tense and the apodosis contains a valid modal verb, thus filtering out other types of conditionals. For this study, I excluded sentences that expressed their conditional value through means other than the grammatical marker "if" (e.g. subject-auxiliary inversion) and sentences containing a modal verb in the protasis (e.g. *If I could go anywhere in the world, I'd go to China.*).

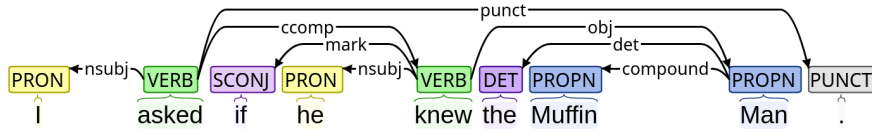


Figure 1: UD parse of a sentence with a non-conditional *if*

Finally, all conditionals that fit all requirements were compiled into a CSV file with the full sentence, the lemma filling the verbal slot in the protasis and metadata about the writing prompt and response, such as the *Englishtown* level the task stemmed from and the topic ID.

2.3 Calculating dispersion across tasks

2.4 Distinctive Collexeme Analysis

2.5 Lexical variability

3 Results

In this section, I outline the results for each of the analyses in the study.

²However, given previous robustness studies of NLP tools on L2 English (Geertzen et al., 2013), similar results would be expected when running the analysis on the original learner texts.

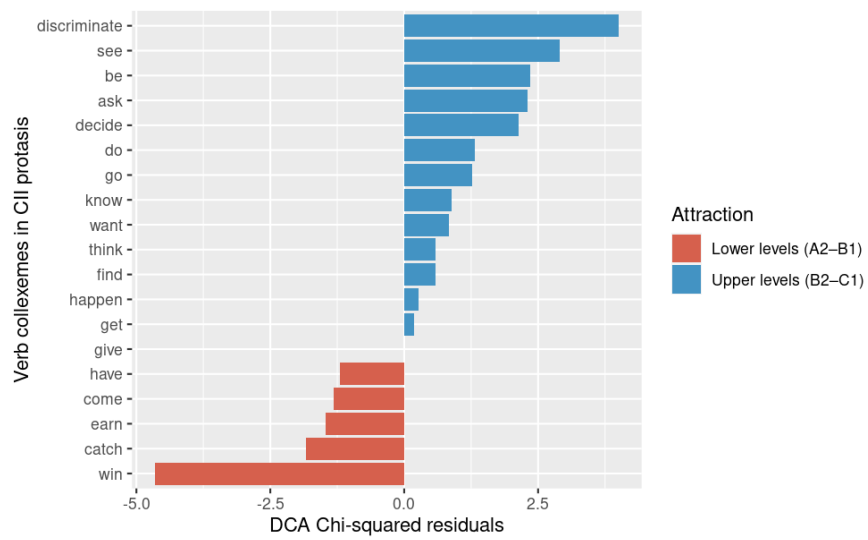


Figure 2: Residuals of the χ^2 -test

4 Discussion

5 Conclusion

References

- Jeroen Geertzen, Theodora Alexopoulou, Anna Korhonen, et al. Automatic linguistic annotation of large scale L2 databases: The EF-Cambridge Open Language Database (EFCAMDAT). In *Proceedings of the 31st Second Language Research Forum. Somerville, MA: Cascadilla Proceedings Project*, pages 240–254, 2013.
- Peng Qi, Yuhao Zhang, Yuhui Zhang, Jason Bolton, and Christopher D Manning. Stanza: A python natural language processing toolkit for many human languages. In *Proceedings of the 58th annual meeting of the association for computational linguistics: system demonstrations*, pages 101–108, 2020.